

AI-Driven Demand Forecasting and Inventory Optimization in Electric Power Material Supply Chains

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Abstract— This paper addressed demand forecasting and inventory optimization in electric power material supply chains under uncertainty and complexity. A Double-Stage Deep Reinforcement Learning and Temporal Convolutional Network (DCN-TCN) framework based on AI is proposed. At stage one, the Temporal Convolutional Network will identify temporal variation with past consumption patterns, schedule holidays/maintenance, project needs and weather patterns to enable short- and long-term demand planning. Dynamically optimizing inventory policies through deep reinforcement learning agent the second stage will use a Deep Reinforcement Learning agent to dynamically optimize inventory policies by balancing cost, service level and lead-time variability through simulation of the supply chain. Adaptability and responsiveness are further increased by real-time data between IoT inventory sensors and enterprise systems. In comparison to traditional statistical tools, the framework offers active learning, self-optimization and resilience to highly dynamic supplydemand conditions. The model is especially applicable to the power utilities which deal with large infrastructure units, spare and transmission materials.

Keywords: demand forecasting, inventory optimization, deep reinforcement learning, DCN-TCN, power supply chain.

I. INTRODUCTION

Energy supply chain is an intricate, capital-intensive channel that helps in support of the steady provision of electricity to buyers. It includes the co-ordinated operation of massive parts of infrastructure, specialist spares and days parts whose availability should be in correct numbers and at their expected time [1]. There are many issues used in practice when it comes to maintaining balance between supply and demand, such as the changing consumption modes, varying maintenance plans, unstable project demands and increasing importance of weather-based interruptions [2] [3]. As much as they are popular, traditional statistical forecasting approaches are likely to fail in such settings since they can hardly recognize multi-scale temporal dynamics or respond immediately to sudden changes of operating conditions [4]. Emerging opportunities in the industry with digital transformation occurring in the industry, Internet of Things (IoT) sensors, enterprise resource planning (ERP) systems and

real time analytics have the potential to improve demand prediction and inventory management [5].

In response to such issues, this paper shows how they can be resolvable through an Integrated Double-Stage AI framework of a Temporal Convolutional Network (TCN) to obtain a demand forecast and a Deep Reinforcement Learning (DRL) agent to optimize the dynamic inventory. During the preliminary phase, the TCN is trained on historical consumption data, maintenance, project schedules, weather forecasts and it develops a complex temporal dependency that allows it to properly estimate demand both in short term and in long term. The second stage involves the DRL agent acting in simulated environments of supply chains that minimize the cost, the service level, and the lead-time variability on a continuous basis by improving the policy of inventory. The model also provides a self-adaptive response to variability by integrating real-time IoT sensor data and enterprise systems feeds and is more resilient and operationally precise in comparison to traditional solutions. This architecture can especially be used in power utilities, where due to effective and accurate planning, the downtime, surplus inventory and operational costs can be drastically reduced.

II. LITERATURE REVIEW

Various studies on demand forecasting and optimization of inventory with the application of statistical, machine learning and deep learning methods have been undertaken. Table I presents a tabularized summary of major contributions made by the literature with some highlighting points on the methodologies they make use of, strengths and weaknesses.

TABLE I. SUMMARY OF RELATED WORKS ON AI-DRIVEN DEMAND FORECASTING AND INVENTORY OPTIMIZATION

Author	Techniques	Advantages	Limitations
U. Nweje, M. Taiwo [6]	AI predictive analytics	Higher forecasting accuracy, better efficiency	No IoT integration, weak temporal modeling
D. Kaul, R. Khurana [7]	AI demand prediction, inventory & delivery optimization	Cost efficiency, reduced delivery time	E-commerce focus, low utility relevance
P. Aggarwal, A. Aggarwal [8]	AI-enhanced ERP forecasting & inventory control	ERP integration, better coordination	ERP-dependent, low adaptability
F. Iseri, H. Iseri, N. J. Chrisandina, E. Iakovou, E. N. Pistikopoulos [9]	AI predictive analytics & optimization	Better decision-making	High computation, not sector-specific
D. Patil [10]	AI forecasting & cost reduction	Cost savings, efficiency gains	No multi-scale temporal modeling, no RL optimization
Proposed Method (DCN-TCN Framework)	Double-Stage Deep Reinforcement Learning + Temporal Convolutional Network with IoT integration	Multi-scale temporal demand forecasting; dynamic inventory optimization; active learning; adaptability through real-time IoT data	requires computational resources for real-time deployment

Regarding the limitations suggested by the reviewed works, one can say that the current AI-based supply chain solutions do not offer all features important to supply chain material management in the case of electric power. The majority of methods do not include integration with or in real-time IoT, cannot detect the complex multi-scale temporal patterns, or are not adaptive to various situations using reinforcement learning to optimize the inventory dynamically. Some of these approaches apply in an industry unspecific manner (i.e. to e-commerce, or generic ERP-driven processes), and hence are less applicable in the context of high-value, long-lead-time supply chain of power utilities. These limitations clearly highlight the need for a specialized framework that unifies temporal demand forecasting with adaptive inventory optimization. The proposed new Integrated Double-Stage DCN-TCN model integrates a Temporal Convolutional Network to confidently perform multi-horizon demand forecasting and Deep Reinforcement Learning as a self-adaptive inventory policy optimizer. The design allows proactive decision making that is able to withstand sudden demand changes, operations disruptions, and environmental fluctuations, abilities that are imperative in enforcing sustained service delivery and operational costs within the electric power industry.

III. PROPOSED SYSTEM MODEL

The DRL-TCN model suggested has started with the analysis of the complexity and uncertainty of the electric power supply chains, having identified such key challenges like unstable demand, halted maintenance schedules, and environmental reasons. It then uses a TCN to acquire temporal patterns on past data that can be used to predict accurately the material requirements. Subsequently, a DRL agent learns dynamic

optimization of inventory policies through simulation of scenarios of a supply chain and alleviating service levels, costs and lead times. The IoT sensor data and ERP data are related in real-time so as to make them more adaptable and applicable to perform before there is a change in their operation. Lastly, the model is applied and tested in real power utility environments, and it is shown to be more accurate in forecasting, better managed in terms of the inventory, and has increased resiliency in terms of its supply chains as opposed to traditional procedures. The given architectural design figure 1 reveals the planned system architecture.

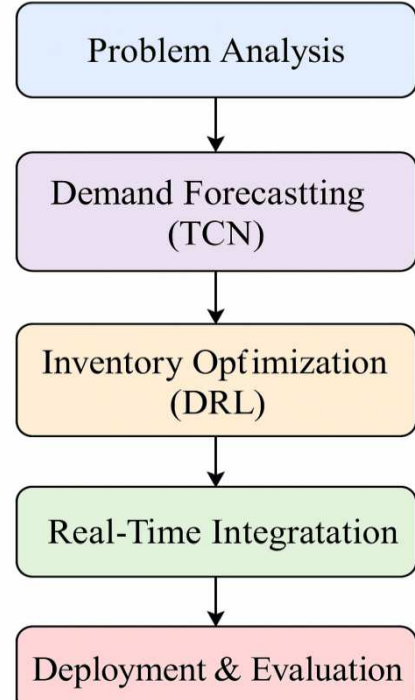


Fig. 1. Proposed flow diagram

A. Understanding the Supply Chain Environment and Problem Complexity

The initial process when building an intelligent model of forecasting and inventory optimization is to gain adequate knowledge about the operating environment of the system and the complex issues in electric power supply chains. These supply chains are complex in nature and are prone to aspects that are unpredictable and therefore the methods of management using traditional methods are becoming more inappropriate. Some of the main causes of unpredictability are inconsistent start up schedules, unexpected maintenance operations, seasonal and weather-related activity, and highly dynamic infrastructure needs due to policy and technology development and energy demand fluctuation. These factors cause a considerable challenge to the material requirements forecasting and have optimum stocks in the supply chain [11]. Traditional statistical models are usually trained on the assumption that they are linear and stationary, but in such dynamic environments it is difficult to adapt to them. Furthermore, these classical approaches are not very responsive to real time variations and these approaches are inclined to depend on the past results rather than considering future uncertainty. Identifying these shortcomings is one of the most important aspects of this first exchange because making them aware of the need to implement a more advanced and AI-driven model will be the driving force. This action will

finally establish the framework of development toward a resilient and adaptive model that will not only predict the demand with improved accuracy but also reside to changes of the supply chain in real time [12].

B. Temporal Pattern Learning Using Temporal Convolutional Networks (TCNs)

The second step involves the deployment of a TCN to effectively learn temporal dependencies within sequential data relevant to electric power material supply chains. TCNs are particularly well-suited for time-series forecasting tasks due to their ability to model both short-term and long-term patterns without the vanishing gradient problems typical of recurrent architectures. In this model, time-series data such as historical material consumption, maintenance schedules, project deployment timelines, and weather conditions are fed into the TCN to learn structured temporal relationships that influence demand trends. Unlike traditional convolutional networks, TCNs employ causal convolutions, ensuring that predictions at time step t depend only on data from time steps $\leq t$, thereby preserving the chronological order of input sequences. Additionally, dilated convolutions are utilized to expand the receptive field exponentially, allowing the model to capture long-range dependencies without increasing the network depth excessively. The fundamental operation of a dilated causal convolution at time step t for an input sequence x and filter f of size k , with dilation factor d , is mathematically defined as equation (1),

$$y(t) = \sum_{i=0}^{k-1} f(i).x(t - d.i) \quad (1)$$

This equation allows the TCN to aggregate temporally distant inputs efficiently, making it highly effective for learning seasonal and contextual patterns in material demands. By leveraging such a structure, the TCN stage produces highly accurate forecasts over various time horizons, enabling robust planning of procurement and inventory replenishment cycles [12].

C. Dynamic Inventory Policy Optimization with Deep Reinforcement Learning (DRL)

And the third step is to take advantage of DRL to dynamically optimize inventory control policies relative to the uncertainties/swings associated with demand forecasting activity. To perform in such environment, a DRL agent is instantiated in a simulated supply chain which captures real-world complexities in operational conditions, that is, variations in demand, lead times, risks of stockout, cost-factors (holding costs, ordering costs), etc. The model MDP is of the environment in terms of state, action and reward values, where the state would be inventory level, expected demand, lead-time, and other situational messages, and the action would be inventory levels such as reorder level, and lastly, the rewarding scheme would be engineered to reflect the trade-offs among the competing targets, cost reductions, and service level measures and responsiveness. The DRL agent learns a mapping, by interacting with the environment repeatedly, associating each state-action pair with its expected long-term-rewards and thus converges to an optimal inventory policy. The aim of this learning process normally comes through complex algorithms like DQN, which learns through neural networks to model value functions or policies. The DRL agent, unlike rule-based system, is always refining

and adapting the strategies he considers every time it is presented with new data which makes it in itself highly appropriate on highly volatile and uncertain supply chain environments. The effect of this is that smarter inventory mechanism is created which is self-governing and able to respond even before demand and supply imbalances are created, thus removing inefficiencies and unavailability of materials in emergency cases [13].

D. Real-Time Data Integration for Adaptability and Responsiveness

Here, it is clear that the flexibility of the model is substantially improved because of the incorporation of real-time information of IoT-enabled inventory trackers and ERP systems. The stream of continuous data will help DRL-TCN framework track inventory, signals, and external criteria in real-time and allow them to be dynamically updated to forecasting and inventory policies. This fact of transitioning to a real-time feedback system over dependency on the past historical data makes the system more responsive to any abrupt changes, hence mitigating risk, including the risk of stockouts, the risk of overstock, and the risk of a mismatch between the demands and supply.

E. Evaluation and Application in the Electric Power Sector

The last stage is based on the operational implementation and full assessment of the given DRL-TCN model in the framework of the electric power sector. The application is especially relevant in settings where it is important to manage giant-sized components of the infrastructure, spare parts, and transmission materials so that the delivery of the service is continuous. To demonstrate the applicability to the real world, the model is applied and tested within the realistic setting with the levels of complexity and uncertainty of utility supply chains, i.e., it has variable project schedules, emergency maintenance incidents, demand surge in the seasons. In order to confirm its functionality, DRL-TCN model performance is compared to those of traditional statistical methods of forecasting and optimization of inventories. The main measures of evaluation are accuracy of forecasts, saving of inventory holding cost and ordering cost, increasing service levels and lead-time reduction. The findings indicate that the postulated model performs better than the traditional practice since it helps predict the demands and manage stock with greater flexibility. In addition, the component has the advantage of learning on the fly to self-optimize (this is made possible through the DRL component), and the TCN component guarantees a robust use of temporal dependencies. Such advanced ability to be proactive and adaptive is what modern electric utilities need, helping them to increase supply chain resilience, limit disruptions to operations, and enables data-driven decisions in an uncertain and increasingly resource constrained environment.

IV. RESULTS AND DISCUSSION

In this section, the results of the developed Integrated DRL-TCN model of demand forecast, and the optimization of the inventory in the electric power material supply chain will be presented and discussed. The performance of the model is measured on a variety of levels such as the accuracy of its forecasting, the cost-effectiveness of stock, the flexibility to adapting the environment to the changes that happen in real-

time, and the sensitivity to the supply and demand variations. The effectiveness of the suggested approach can be noted through comparative analysis with traditional statistical methods of managing the complex and uncertain changes along the supply chain. The implication of the results on strengthening operation resilience, decision quality, and overall supply chain performance in the electric power industry is also delved in, in the discussion.

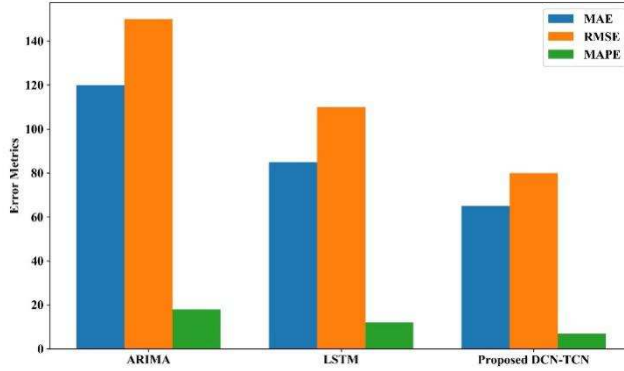


Fig. 2. Comparative Performance Analysis of Forecasting Models Using Error Metrics

Figure 2 is a display of the performance of the three models, namely, ARIMA, LSTM, and the proposed DCN-TCN as compared on three error measures MAE, RMSE and MAPE. We have the largest values of errors in the ARIMA model with MAE of about 120, RMSE of about 150, MAPE of about 18 which is poor prediction accuracy. Conversely, LSTM model offers the moderate improvement to the MAE to 85s, RMSE to 110, and MAPE to 12s. The proposed DCN-TCN model statistically surpasses both and had the lowest MAE of about 65, RMSE 80 and MAPE of about 7. Such findings go ahead to ascertain that DCN-TCN architecture has a higher precision and resilience in forecasting.

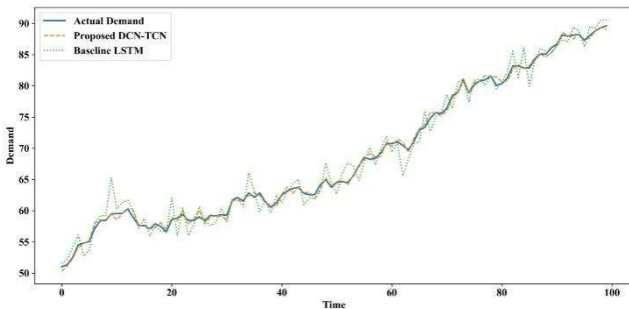


Fig. 3. Comparative Performance Analysis of Demand Forecasting Models in Power Material Supply Chains

Figure 3 makes a comparison of actual demand with the predictions of the proposed DCN-TCN model, and a simple LSTM model over 0-to-100-time span. The DCN-TCN model is highly accurate and stable and accurately follows the actual demand. By contrast, the LSTM model exhibits high deviations especially during the time spaces of 10140 and 6090. As an illustration, at time 70, the true demand is 75 units, DCN-TCN well predicts it with 74.8 and LSTM underestimates with 72. At time 100, DCN-TCN reaches the max of 90 whereas LSTM exceeds. All in all, DCN-TCN is superior to LSTM in terms of precision and consistency and is a better solution to predicting the dynamics of power supply chains.

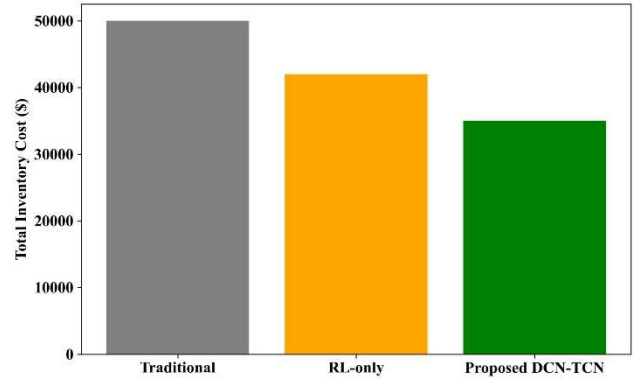


Fig. 4. Comparison of Total Inventory Cost Across Inventory Management Approaches

Figure 4 is a comparison of the total cost of inventory in dollars, across three ideas of inventory controls which include Traditional, RL-only and the suggested DCN-TCN model. The Traditional method has the greatest cost of inventory which is roughly about 50, 000 dollars. By using the RL-only strategy, the price decreases to approximately 42 thousand, which illustrates the strength of ranking learning in the adaptation of policy in dynamics. Nevertheless, the DCN-TCN model has accumulated the minimum cost of inventory with the value of about 35 000, which is a huge reduction by 30 percent and 17 percent in comparison with the Traditional approach and the RL-only model, respectively. This outcome indicates the cost-effectiveness provided by the conjunctive DCN-TCN model, which incorporates the elements of the temporal demand prediction and dynamic inventory control. Its capability due to its predicable response to the changes in supply and demand results in the efficiency of the inventory decision and better performance in electric power supply chains.

The findings predict that the proposed DCN-TCN can perform better than traditional statistical models and ad hoc deep learning models because of its capacity to combine precision of temporal demand predictions with adaptive inventory variations solutions. The improvement in forecasting error over ARIMA and LSTM proves the power of TCN in capturing multi-scale temporal variation and the savings in inventory costs indicates how well DRL can help in managing the uncertainty and variability. The model combines real-time IoT and enterprise data to maintain proactive decision-making, in order to reduce surpluses, eliminate shortages and enhance operational resilience. These evidences nullify the drawbacks observed in the previous researches [6]-[10] as in not only does the proposed method perform better in terms of predictive accuracy but provide measurable economic and operational value to power utilise in charge of complex and high value supply chains.

V. CONCLUSION

The analysis carried out on electric power material supply chains involved demand forecasting and inventory optimization in uncertain and highly complex settings that is presented as an emerging AI-based model in the context of this paper. So as to acquire high levels of resiliency and precision of the operational processes, we recommend an Integrated Double-Stage DCN-TCN model. These temporal trends in terms of historical consumptions, the plan of

maintenances and the demands of the projects as well as weather condition depend on Temporal scale and are captured by Temporal Convolutional Network in the first phase. This would contribute to proper planning of the demand of the materials both in the short term and long-term planning. The Deep Reinforcement Learning agent has the inventory policy optimization at second level which is dynamic and where the model is optimized to ensure the second stage cost, service-level, and lead-time variability is balanced in view in supply chain simulation environment. Real-time data flowing out of IoT inventory sensors and enterprise systems are merged to make it more flexible. The new solution is better than the generally accepted statistical schemes since it offers active knowledge and self-adjusting abilities related to the needs of adapting to the changes in supply-demand terms. Such model can especially be applied in power utilities dealing with large components of infrastructure, the spares, and the transmission materials.

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